

Losing Face but Winning Fame: Coopetitive Social Comparison and Streamer

Performance in Livestreaming Platforms

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Abstract

Digital platforms increasingly deploy competitive features to drive user engagement. This study investigates player knockout (PK) events, livestreaming contests in which two streamers compete on a split screen for audience gifts. We conceptualize PK as a form of coopetitive social comparison: social comparison between streamers that is simultaneously competitive and cooperative, whose outcome is determined not only by the streamers themselves but also by the audience through collective gifting. The empirical tension is that players in PK could win or lose on the surface, yet they might have differential and consequential outcomes in terms of user engagement, follower flow, and financial performance. Using proprietary data from a leading livestreaming platform, we conduct two empirical studies. Study 1 employs Double Machine Learning over 500,000 livestreaming sessions and shows that PK participation increases total gifting by over 300%, average viewer watch time by about 106%, and new follower acquisition by over 130%. A decomposition of total gifting shows that PK increases viewer count and average gifting per gifter without affecting the gifting conversion rate, and a mediation analysis of viewer comments shows that competition awareness is a plausible underlying mechanism in the effects of PK. Study 2 leverages over 130,000 Random PK events to examine how follower-size comparison and content-category alignment shape streamer outcomes. Our results show that PK outcomes can diverge across dimensions: facing a bigger opponent lowers the focal streamer's likelihood of surface winning but increases gifting and net follower growth, while same-category matching increases gifting and bidirectional follower attraction without affecting win probability. A viewer-source decomposition further shows that the focal streamer's own viewers give more under both

structures, with additional gifting coming from viewers outside either pre-PK audience against a bigger opponent and from the opponent's viewers under same-category matching. This study extends social comparison theory by showing that third-party audiences can shift from passive observers to active participants whose collective behavior determines the comparison outcome. It further broadens comparison structure beyond comparison direction to include content-category alignment. This work also advances coopetition research by showing that competition and mutual gains can coexist in PK and that comparison structure shapes the audience responses through which these coopetitive outcomes unfold. Our findings offer practical implications for using PK as an effective gamification design to mobilize collective effort in livestreaming.

Keywords: Social Comparison, Player Knockout, Coopetition, Gamification, Livestreaming, Streamer Performance, User Engagement, Platform Monetization

1. INTRODUCTION

Livestreaming platforms, such as Twitch and TikTok, rely on user engagement to sustain operations and to enable monetization (Zhao et al., 2025). Over one-fourth of Internet users watch livestreaming weekly, and this massive user base is driving the livestreaming market’s projected growth from about \$100 billion in 2025 to over \$340 billion by 2030 (Kumar, 2025). The extant literature has explored a range of strategies for driving user engagement in livestreaming platforms, such as leveraging liveness to strengthen viewers’ sense of presence and connection (Duani et al., 2026), cultivating positive streamer affect to stimulate likes, chats, and virtual gifting (Lin et al., 2021), encouraging content-category switching to generate viewer-flow spillovers across channels (Zhao et al., 2023), and deploying human and bot moderators to mitigate the engagement-dampening effects of audience growth (Zhao et al., 2025).

More recently, platform-mediated, interactive, public contests, such as Twitch Rivals and TikTok Live, have emerged as a form of platform design for user engagement. In this work, we focus on an increasingly prominent design, known as player knockout (PK) events, in which two streamers appear side-by-side on a split screen and compete in a time-limited, head-to-head contest. The winner is determined by the total value of virtual gifts received from viewers during the event, making the audience the arbiter of the contest outcome.

Drawing on Social Comparison Theory, which concerns individuals evaluating their abilities, opinions, and social standing by comparing themselves with others (Garcia et al., 2013; Malhotra, 2010), as well as Coopetition Theory, which studies the simultaneous presence of competition and cooperation between the same actors (Hoffmann et al., 2018), we conceptualize PK as a form of *coopetitive social comparison*, for two reasons. First, unlike conventional social comparisons such as leaderboards and peer-performance displays, where audiences remain largely passive observers, PK involves the audience as active participants whose collective

gifting determines the outcome. Second, PK is structurally coopetitive. Its competitive dimension arises because the two streamers vie for victory, and viewers' gifts determine who wins. Its cooperative dimension arises because the streamers jointly produce a split-screen performance that exposes each to the other's audience and can generate attention, gifting, and follower gains for both. These dimensions coexist in every PK, while the comparison structure shapes how they are expressed in audience responses. Together, these two features make PK a social comparison between streamers that is simultaneously competitive and cooperative.

This work aims to understand how PK events, as a form of coopetitive social comparison, influence streamer performance on livestreaming platforms. Extant livestreaming research has focused primarily on streamers' financial performance, such as viewer gifting outcomes (Lin et al., 2021; Lu et al., 2021; Yao et al., 2025). However, in addition to financial outcomes, PK may lead to broader consequences by intensifying viewer engagement, exposing each streamer to the other's audience, and influencing follower acquisition and attrition. Followers are important for streamers because a growing follower base increases a streamer's visibility, builds popularity, and creates greater opportunities for sustained engagement and future monetization (Nistor et al., 2025). Accordingly, beyond the surface win-lose result, it is important to understand how PK events translate into less visible but consequential outcomes for streamers: financial returns, audience engagement, and follower acquisition and attrition.

Bearing the above in mind, we seek to answer the following research questions: *(1) What are the effects of participating in PK events on streamer performance (e.g., gifting revenue, viewer engagement, follower change)? (2) How do PK structures, namely follower-size comparison and content-category alignment, shape streamer outcomes? (3) What could be the plausible underlying mechanisms?*

To answer our research questions, we collaborate with a leading online livestreaming platform in Asia (hereafter, the partner platform) and analyze proprietary data from this platform across two empirical studies. In Study 1, we analyze a dataset of 518,690 livestreaming sessions and employ Double Machine Learning (DML) to estimate the effects of PK participation on streamer performance. The results show that participating in PK events significantly enhances streamer performance, with more than a 300% increase in total gifting, about a 106% increase in average viewer watch time, and an increase of over 130% in new follower acquisition. A decomposition of total gifting shows that PK increases viewer count and average gifting per gifter with no statistically detectable change in the gifting conversion rate.¹ Our further mediation analysis of viewer comments suggests a significant mediating role of competition awareness in PK, reflected in comments referring to the contest. Moreover, we find that the effects of PK participation also extend beyond the PK itself: streamers who participate in a PK recover more previously inactive viewers during the session, are more likely to use the feature again within 30 days and to change what and when they stream next, and gain views over the following 30 days, although their loyal-fan growth over that horizon does not rise.

In Study 2, we focus on the platform’s Random PK pairings, in which the platform rather than either streamer chooses the opponent, providing a quasi-experimental source of variation in comparison structure. Specifically, we examine how upward comparison (comparing with an opponent with a larger follower size) and same-category alignment (comparing with an opponent in the same content category) affect streamer outcomes. Our findings show that PK outcomes vary by comparison structure. Facing a bigger opponent lowers the focal streamer’s likelihood of winning but increases gifting and net follower growth, while same-category matching increases

¹ The gifter-to-viewer ratio shows no statistically detectable change, which provides no evidence for a herding pattern operating through the conversion of viewers into gifters.

gifting and bidirectional follower attraction with no statistically detectable change in win probability. Under both structures the focal streamer's own viewers give more, and a bigger opponent also draws gifting from viewers who were in neither streamer's room beforehand, whereas a same-category opponent's own viewers give to the focal streamer. Together, these results suggest that the focal streamer's existing supporters respond competitively in both types of matchups, while the additional gifting comes from different audience segments depending on the comparison structure. Our heterogeneity analysis further shows that the gifting gain from facing a bigger opponent is larger for small streamers than for big ones, and that facing a same-category opponent raises gifting for small streamers but not for big ones. Beyond the PK itself, facing a bigger opponent raises the number of viewers who go on to follow both streamers and the number of returning viewers in the focal session, and it is followed by higher views and loyal-fan gains in the streamer's subsequent Random PK sessions over the following 30 days, whereas facing a same-category opponent reduces returning viewers and is followed by higher views in those sessions.

This work makes several contributions to related literature. First, we extend social comparison theory by showing that a comparison can be audience-activated and that comparison structure shapes the audience responses and streamer outcomes it produces. Prior work treats social comparison as a comparer-centered process: the person who receives comparison information evaluates themselves and adjusts their own behavior, while third parties remain observers (Festinger, 1954; Krasnova et al., 2015; Li et al., 2021; Shi et al., 2025). In PK, the comparison between two streamers triggers a compare-choose-support chain in the audience, and the outcome is settled by viewers' collective gifting rather than by the comparers alone; viewers' comments during the contest reflect a pattern consistent with the competitive aspect of

coopetitive social comparison. We further broaden comparison structure beyond comparison direction, the upward-versus-downward distinction that organizes existing work, to content-category alignment, a dimension of potential audience overlap and sharing. Alignment raises gifting and follower attraction in both directions with no statistically detectable change in win probability. Where prior work shows that comparison direction shapes how the comparer responds, comparison structure in PK also determines who else responds, with direct consequences for gifting and follower dynamics in the creator economy.

Second, we advance coopetition research by showing that competition and mutual gains can coexist in PK, and that comparison structure shapes the audience responses through which these coopetitive outcomes unfold. We show that the relative win-lose result can coexist with absolute financial gains for both participants (Study 1) and an exchange of followers between them (Study 2), so coopetitive social comparison cannot be evaluated through a single performance lens. We further show that comparison structure shapes which audiences carry these gains. The focal streamer's own supporters give more under both structures, consistent with the competitive dimension, while the second gain, consistent with the cooperative dimension, comes from a different audience under each structure: from viewers outside either pre-PK audience when the opponent has more followers, and from the opponent's own viewers when the opponent streams in the same category. In doing so, we bring an individual-level theory of how people react to social comparison into a coopetition literature built on organizations such as firms, and use it to explain the process through which a contest that only one side can win yields gains for both.

Third, we contribute to the gamification literature by identifying PK as an effective gamified design deployed in a synchronous, collective setting. Gamification research has largely

studied one element at a time (a badge, a point system, a leaderboard, or a competitive matchup) and has measured its effect on the person who earns the badge or climbs the board, typically in asynchronous settings such as shopping, knowledge sharing, and training (Ho et al., 2023; Liu et al., 2017; Santhanam et al., 2016; Wang et al., 2026). PK combines real-time competition, dynamic scoring, time pressure, audience participation, and reward-punishment in a gamified design that involves both competitors and their audiences. Such a design raises the number of viewers and the amount each gifter gives, and within Random PK a bigger opponent draws gifting from viewers outside either pre-PK audience and brings gains to a streamer who is less likely to win the visible contest. In rank-based designs the benefit accrues to those who rank the highest, and falling behind can discourage the weaker party (Yang et al., 2023); PK shows that cooperative social comparison can produce gains for both competitors. PK thus shifts the function of gamification from motivating an isolated individual to mobilizing an audience whose collective engagement can help both contestants.

Our results also offer practical implications. For platforms, PK is an effective lever for engagement and monetization that works through the audience: sessions with a PK draw more viewers and larger gifts per gifter, and within Random PK, part of the gifting response to comparison structure comes from viewers who were not in the streamer's room before the PK. Matching is where the design choices lie. In any pair only one streamer faces a bigger opponent, and the gifting gain from that condition is larger for small streamers, so a platform can allocate upward comparisons to small streamers, for whom same-category pairings raise gifting as well, while using incentives to bring big streamers into these matchups. For streamers, a matchup they are less likely to win is not costly in absolute terms: a small streamer who faces a bigger opponent receives more gifts and ends the session with more followers, and a same-category

opponent also raises a small streamer's gifts, so both matchups may be beneficial. For the design of competitive features more broadly, PK shows that a contest decided by the audience turns spectators into participants and can be built so that both competitors gain, which sets it apart from rank-based designs in which one competitor's rise is another's fall.

2. THEORETICAL FOUNDATION AND RELATED LITERATURE

2.1 Social Comparison Theory

Social comparison theory posits that individuals are inherently motivated to compare with others and minimize perceived discrepancies between themselves and others (Festinger, 1954). This fundamental psychological process drives competitive behaviors aimed at reducing perceived discrepancies and preserving or enhancing one's perceived superiority (Garcia et al., 2013; Tauer and Harackiewicz, 1999). Prior research has identified two main directions of social comparison: upward comparison, where individuals compare themselves to superior others, often fueling the desire to compete and outperform (Malhotra, 2010; Tesser, 1988); and downward comparison, which can also elicit competitive responses, especially when those performing worse are seen as potential threats to one's current advantage (Garcia and Miller, 2007; Solnick and Hemenway, 1998). Competitive rivalry extends these dynamics further, as repeated comparison with the same target creates relational stakes in personal pride, reputational standing, and emotional investment in outcomes (Goethals and Darley, 1977; Kilduff et al., 2010; Milstein et al., 2022).

Information Systems scholars have employed social comparison theory in various contexts for understanding user behaviors, such as group interactions in the adoption of collaborative technologies (Kang et al., 2012), negative emotions like envy on social networking sites (Krasnova et al., 2015), and the motivational appeal of competitive features in gamified systems (Liu et al., 2017). The extant literature has largely focused on how comparison affects individuals' own behaviors, evaluations, or emotional states. For example, Li et al. (2021) find

that peer performance information motivates men more than women in competitive learning, due to men's stronger social comparison orientation. Similarly, Shi et al. (2025) show that popularity feedback in online dating prompts strategic adaptations, especially among men, who adjust their date-seeking behavior in response to comparison-based information. Across these studies, even when comparison information is publicly displayed, its behavioral consequences remain predominantly comparer-centered: the individual receiving the comparison information evaluates themselves and adjusts their own behavior, while third-party audiences remain largely passive observers.

Our work extends social comparison theory to the context of livestreaming, particularly PK events. PK events pair streamers in public, time-limited contests, wherein streamers perform live to compete for viewer support through virtual gifting, and the outcome is determined by the total value of gifts each streamer receives from viewers. This process differs from traditional social comparison in one important respect: the outcome of the comparison is determined not solely by the comparers but by the audience, whose collective gifting adjudicates which comparer wins. The comparison between streamers triggers a behavioral chain in which audience members compare the two, choose whom to support, and express that support through gifting, watching, and following. The comparison outcome is therefore produced by the audience's collective response rather than only by the comparers' own behavior, a pathway that comparer-centered social comparison research does not theorize (Festinger, 1954; Garcia et al., 2013).

Beyond the question of whether comparison can be audience-activated, a second question arises: what role does comparison structure play in this process? Traditional social comparison research has long recognized that comparison structure matters, distinguishing upward from downward comparison and examining how each direction shapes the comparer's self-evaluation

and motivation (Garcia et al., 2013; Malhotra, 2010). However, the broader role of comparison structure, particularly in shaping third-party audience responses and outcomes beyond the comparer's own adjustment, remains understudied. In the PK context, two structures are central. Follower-size comparison captures comparison direction: facing a larger opponent constitutes upward comparison and may lead the focal streamer's supporters to give more. Content-category alignment captures audience preference overlap: pairing with a same-category opponent may facilitate cross-audience discovery and follower movement. By examining both structures, our study broadens the theory beyond comparison direction and shows how these structural dimensions shape audience responses and produce different patterns of streamer outcomes.

2.2 Coopetition Theory

Coopetition theory studies the simultaneous presence of competition and cooperation between the same actors (Brandenburger and Nalebuff, 1996; Hoffmann et al., 2018). Competition is the pursuit of one's own interest at the potential expense of others; cooperation is the joint pursuit of mutual interests that neither party could achieve alone (Das and Teng, 2000). These two logics are contradictory: cooperation demands openness and sharing, whereas competition demands protection and private maximization. When they operate together, the result is a tension that cannot be eliminated, only managed (Larsson et al., 1998; Zeng and Chen, 2003). A central question in coopetitive systems is therefore not simply whether competition and cooperation coexist, but what shapes how each is expressed.

The extant coopetition literature has predominantly studied this tension among organizations, particularly firms. Across this literature, the coopeting actor is typically the organization (Hoffmann et al., 2018; Larsson et al., 1998; Mathias et al., 2018), and tension is managed through governance mechanisms such as formal contracts, relational norms, collective

identity, and organizational structure (Gulati and Singh, 1998). Outcomes are also generally determined through the strategic choices and actions of the coopeting parties themselves (Hamel, 1991). Tsai (2002) extends this perspective to coopetition among business units within a multiunit corporation but continues to focus on organizational subunits. Although prior research has examined how competition and cooperation coexist in coopetitive relationships (Hoffmann et al., 2018), less is known about how specific features of the interaction structure shape the ways in which these two dimensions are expressed. Existing studies point to factors such as collective identity (Mathias et al., 2018) and formal organizational structure (Tsai, 2002), but the role of comparison structure in shaping the behavioral responses through which coopetitive outcomes emerge remains underexplored.

PK provides a setting in which competition and cooperation are structurally intertwined. The two streamers compete because each pursues victory and the win-lose outcome is determined by their relative gifting scores. At the same time, they cooperate by co-producing a joint split-screen performance that can attract and engage audiences for both parties. Our study examines comparison structure as a key factor shaping how competition and cooperation are expressed in PK events. In the PK context, coopetitive social comparison can be structured along two dimensions: follower-size comparison and content-category alignment. Facing a bigger opponent may intensify support from the focal streamer's existing audience, while the joint contest may also attract engagement from viewers beyond the focal streamer's own audience. Same-category alignment may similarly strengthen support from existing viewers while facilitating cross-audience discovery among viewers with shared content preferences. Our study thus connects coopetition theory to social comparison theory by showing that comparison structure shapes the audience responses through which coopetitive outcomes unfold. Because

platforms can configure these comparison structures through their matching design, they can also influence how competition and cooperation are expressed in PK events.

Coopetitive social comparison also produces outcomes that cannot be characterized in a single direction. On the one hand, the comparison outcome is competitive: one streamer wins and the other loses according to their relative gifting scores. On the other hand, both streamers may realize absolute gains, as the joint performance can expand audience engagement and stimulate gifting for both parties. PK can also reshape streamers' follower bases by generating follower acquisition and attrition, creating another dimension along which gains and losses may coexist. By revealing these divergent outcomes, our study shows that the consequences of PK events cannot be evaluated through a single performance outcome.

2.3 Gamification in Livestreaming

Gamification is an established design approach in IS research, and Liu et al. (2017) provide a framework for designing and studying gamified information systems around meaningful engagement. Much of the empirical evidence concerns one element at a time and the behavior of the individual user that element targets. Badges raise activity in a peer-to-peer trading service in a field experiment (Hamari, 2017), and on Stack Overflow a badge system that offers more badges, more badge categories, and higher badge levels raises knowledge contributions, with a larger effect on quality than on quantity (Wang et al., 2026). Status hierarchies motivate contributions as users approach the next rank, but effort drops once the rank is reached and the effect shrinks at higher ranks (Goes et al., 2016). In a randomized field experiment at a shopping mall, badges and leaderboards raise sales by 21.5% and 22.5%, respectively, with effects that persist after the elements are removed (Ho et al., 2023). These settings are asynchronous and

individual: users pursue their own purchases, answers, or trades, and the game element acts on each user's private motivation.

Competition is the game element closest to PK, and research on it shows that its effects depend on the structure of the matchup. Players who compete against opponents of similar skill play more games and play longer (Liu et al., 2013). In gamified training, competing against a weaker opponent improves trainees' self-efficacy and learning, while competing against an equal raises their engagement, so the matchup that best serves one outcome is not the one that best serves another (Santhanam et al., 2016). In an online exercise competition, rankings, performance gaps, and rivalry intensity move active, moderate, and inactive exercisers differently, and a deteriorating performance discourages inactive exercisers (Yang et al., 2023). Across these studies the unit of analysis is the competitor: the design changes how hard the ranked or matched individual works, and third parties, when present, are the audience to whom ranks are displayed rather than actors whose behavior decides the outcome. One exception gamifies the givers rather than the fundraiser: a top-donor leaderboard on a medical crowdfunding site brings in more engaged donors and more total funds, while the amount given per donor falls (Stratopoulos and Ye, 2024).

The livestreaming literature has grown rapidly (Gu et al., 2024; Zhao et al., 2025). Studies of viewer gifting have explained it through emotional contagion from the streamer (Lin et al., 2021), social image concerns and herding (Lu et al., 2021), status competition among viewers (Yao et al., 2025), and viewer-to-viewer interaction in the comment stream, where the presence of others, debate among viewers, and emotional expressions all raise paid gifting (Zhou et al., 2018). Gamified features are gaining popularity in this setting. Streamers gamify their own broadcasts with competitions, targets, and games played with the audience (Johnson and

Woodcock, 2019), and viewers' social motivations for engaging, including social interaction and a sense of community, predict their support of the streamer (Hilvert-Bruce et al., 2018). These studies provide important insights into viewer behavior within livestreaming environments, but say little about how platform-designed gamified structures shape collective audience engagement, and systematic evidence on competitive gamification in this setting remains limited. More broadly, gamification research has predominantly examined discrete game elements and their effects on individual users, rather than integrated designs that organize real-time interactions among multiple participants.

As a platform-designed intervention, PK combines real-time competition, dynamic scoring, time pressure, audience participation, and reward-punishment into an integrated experience. More importantly, PK involves gamification in a synchronous, collective setting: streamers jointly produce the contest, while audience members participate through gifting and their collective behavior determines the competitive outcome. Our work therefore extends gamification research from individual responses to collective participation in an integrated, synchronous livestreaming environment, and shows how a platform-designed feature can shape engagement and monetization through cooperative social comparison.

3. RESEARCH METHODOLOGY

3.1 Research Context

Our research setting is the partner platform, a leading online livestreaming platform in Asia that operates at a large scale, with hundreds of millions of monthly active users, and functions as a two-sided market, facilitating livestreaming interactions for streamers and their audiences. The partner platform encompasses a diverse range of content, including entertainment, gaming, cooking, lifestyle, and more. During livestreaming sessions, streamers engage with viewers through real-time chat, comments, and virtual gifts. These virtual gifts, which can be converted

into real currency, provide streamers with a source of financial support. The partner platform retains a commission fee during the conversion process. This voluntary payment approach, where viewers offer monetary contributions to streamers at their discretion, aligns with the Pay-What-You-Want (PWYW) model documented in the literature (Lu et al., 2021; Yao et al., 2025).

Our study focuses on the player knockout (PK) events, an interactive gamified design in which streamers engage in real-time performance-based contests during livestreaming sessions. In a typical five-minute PK event, two streamers appear on a split screen (Figure 1) and perform activities, such as singing, dancing, or playing games, to stimulate viewer engagement and gift contributions.² Viewers can actively participate in the PK events by sending virtual gifts to support their preferred streamer, directly influencing the contest's outcome. At the end of the PK event, the streamer with the greater gifting amount 'wins,' and the losing party undergoes a figurative punishment, such as singing a song chosen by the winner. These non-monetary stakes add a public, performative cost to losing, which may further heighten competitive effort and audience engagement during the contest. Additional institutional details regarding the PK interface, viewer participation, information visibility, matching procedures, and winner determination are provided in Online Appendix A.

The PK events on the partner platform occur in two ways: Invited PK, where one streamer invites another, and Random PK, which is the platform's term for its algorithmic matching system.³ In Invited PK events, a streamer selects a specific opponent, contingent on the opponent's acceptance. In Random PK events, all streamers who click the Random PK button within a short time window are pooled together. The platform then assigns PK pairs using a

² PK events can involve more than two streamers. We focus on two-streamer PK events in this study as they are the most common ones on the platform.

³ The platform introduced PK in 2018, well before our observation window opens in January 2022, so the sessions we study come from a mature rather than a newly launched feature.

proprietary algorithm that resembles a stratified matching process based on streamers' characteristics. The user interface (UI) of the PK events remains consistent across the two PK approaches. We empirically investigate both PK formats. Study 1 compares sessions with and without a PK event; its treated sessions include both Invited and Random PK, and Online Appendix F estimates the two formats in a single specification against the same no-PK baseline. In Study 2, by leveraging the platform-generated variation in the structure of cooperative social comparison among Random PK events, we explore how comparison structure influences the focal streamer's performance and decompose the gifting response by viewer source.

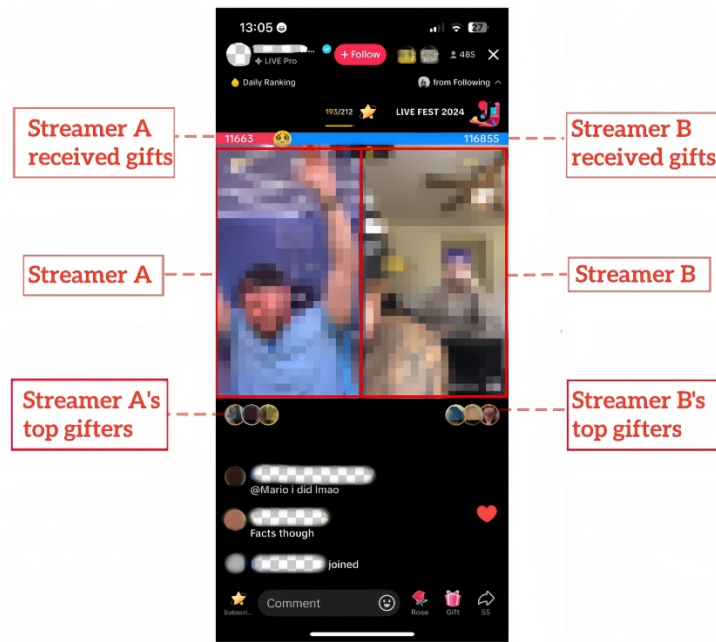


Figure 1: Screenshot of the User Interface During a PK Event in Livestreaming

3.2 Data and Measures

We obtain proprietary data from the partner platform. We describe the data and variables for each study separately below.

3.2.1 Dataset for Study 1: With vs. Without PK

For Study 1, we retrieve data at the livestreaming session level, focusing on each active streamer’s first eligible session of the day, with each session containing no more than one PK event.⁴ Our observation window spans January 1, 2022, to September 1, 2024. To achieve computational feasibility, we randomly sample 1% of eligible active streamers and retain all of their first-session-per-day observations, yielding a final sample of 518,690 sessions from 3,485 unique streamers.

The independent variable of Study 1 is *With PK*, a binary indicator representing whether a livestreaming session includes a PK event. The dependent variables of Study 1 measure several dimensions of streamer performance, including streamer revenue, measured by the total value of gifts that a streamer received during a livestreaming session; viewer engagement, measured by the average viewing duration per viewer during a livestreaming session; audience growth, measured by the number of new followers and net follower gains; fan commitment, measured by the number of new loyal fans who joined the streamer’s fan group; audience interactions, measured by the number of follower connections formed between viewers during the observation live session; and audience churn, measured by the number of existing followers lost.

In addition, our dataset includes observations on streamer characteristics, such as gender, age, streamer category, region, city level, community type, influencer account status, streamer tenure, follower count, loyal fan count, and following count. It also contains streaming device usage data, including device type, app version, and device price range. We further incorporate livestreaming session features, such as live session content category and live session start hour. We also account for prior viewer engagement, measured over the 10 days preceding the observed

⁴ We define an active streamer as one with at least one historical live session during the week preceding the observed session.

livestreaming session; these engagement metrics include cumulative livestreaming duration, total comments, likes, unique commenters, voice comments, shares, and the number of unique users who shared content during livestreaming sessions. We additionally include historical PK experiences, comprising historical PK count, historical PK win count, and historical PK rejections. Finally, we incorporate historical revenue data, captured as the total value of gifts received during the same 10-day period prior to the session.

Further, we include a set of variables used to examine where the gifting effect comes from and what viewers respond to. To decompose the gifting effect, we construct *Viewer Count*, *Gifter Count*, *Conversion Rate* (defined as the ratio of *Gifter Count* to *Viewer Count*), and *Average Gifting per Gifter*. To characterize what viewers express during the focal comment window (the 5-minute window defined in Section 3.3.1), we use a dictionary-based approach to construct three discursive-engagement measures: *Competition Awareness*, *Emotional Exclamation*, and *Relationship Building*. Each measure captures the session-level occurrence of category-specific keywords in the original Chinese comments. We use a large language model (LLM) to categorize candidate keywords in representative comment contexts and then manually review and refine the resulting dictionaries. Only keywords judged to be clear indicators of one of the three theoretically defined categories are retained.⁵

Table 1 presents the variables and descriptions for Study 1, and we report the descriptive statistics of these variables in Table 2.⁶ Unless otherwise noted, all continuous dependent, control, mediator, and composition variables are log-transformed to reduce distributional

⁵ The theoretical grounding of the three measures, the keyword dictionaries, and the construction procedure are detailed in Online Appendix D.

⁶ All continuous variables referenced above are desensitized by the partner platform for data sensitivity reasons: each variable is scaled by a constant factor. Because each factor is constant across observations, the scaling shifts intercepts and leaves the log-point estimates approximately unchanged; Online Appendix B gives the derivation and its limits.

skewness, while categorical variables are encoded using dummy indicators. Percentages reported in the results section are calculated as $\exp(\beta) - 1$ for log-transformed outcomes.

Table 1: Variable Descriptions for Study 1: With vs. Without PK

Variables	Descriptions
Independent Variables:	
With PK	Indicates whether the observation live session has a PK event (0 = without a PK event, 1 = with a PK event).
Dependent Variables:	
Total Gifting	Total monetary value of gifts received during the observation live session.
Avg Play Duration per Viewer	Average play duration per viewer during the observation live session.
New Follower Gain	Number of new followers a streamer gains during the observation live session.
Existing Follower Loss	Number of existing followers a streamer loses during the observation live session.
Net Follower Gain	Net change in the number of followers during the observation live session. Because the net change can be negative, it is shifted by its sample minimum before the log transformation, $\log(y - y_{\min} + 1)$.
Viewer-to-viewer Follower Gain	Number of times viewers follow other viewers during the observation live session.
New Loyal Fan Gain	Number of new members joining a streamer's fan group during the observation live session.
Control Variables:	
Gender	Gender of the streamer (0 = Male; 1 = Female).
Age Category	Age category of the streamer (3 = 18–23; 4 = 24–30; 5 = 31–40; 6 = 41–49; 7 = 50 and above).
Streamer Content Category	Content category of the streamer (1 = Expert; 2 = E-commerce; 3 = Show; 4 = Game).
Region	Region of the streamer (0 = Unknown; 1 = Northern; 2 = Southern).
City Level	Tier of the streamer's city (0 = Unknown; 1 = First-tier; 2 = New first-tier; 3 = Second-tier; 4 = Third-tier; 5 = Fourth-tier; 6 = Fifth-tier).
Device Type	Type of device used (1 = Android; 2 = iOS).
App Version	App version used (0 = Unknown; 1 = Lite; 2 = Express; 3 = Regular).
Device Price Range	Device price range (0 = Budget; 1 = Mid-range; 2 = High-end; 3 = Premium).
Community Type	Community classification (0 = Unknown; 1 = Urban; 2 = Foreign; 3 = Rural; 4 = Township).
Influencer Account	Indicates if the streamer is a verified influencer (0 = No; 1 = Yes).
Live Session Category	Content category of the observation live session, coded 0 to 13; code 0 denotes an unknown category and is retained as its own category. The platform does not permit disclosure of the category labels.
Live Session Start Hour	Starting hour of the observation live session.
Tenure Days	Number of days since registration (streamer tenure).
Follower Count	Total follower count before the observation live session.
Following Count	Number of users followed by the streamer before the observation live session.
Loyal Fan Count	Count of loyal fans before the observation live session.
Historical Total Gifting	Total gift value in live sessions over the 10 days before the observation live session.
Historical Play Duration	Total play duration in live sessions over the 10 days before the observation live session.
Historical View Count	Total number of views in live sessions over the 10 days before the observation live session.
Historical Viewer Count	Total viewer count in live sessions over the 10 days before the observation live session.
Historical Like Count	Total like count in live sessions over the 10 days before the observation live session.
Historical Like Users	Number of unique users who liked content in live sessions over the 10 days before the observation live session.

Variables	Descriptions
Historical Comment Count	Total comment count in live sessions over the 10 days before the observation live session.
Historical Commenters	Number of unique commenters in live sessions over the 10 days before the observation live session.
Historical Voice Comments	Total voice comment count in live sessions over the 10 days before the observation live session.
Historical Voice Commenters	Number of unique voice commenters in live sessions over the 10 days before the observation live session.
Historical Share Count	Total share count in live sessions over the 10 days before the observation live session.
Historical Share Users	Number of unique users who shared content in live sessions over the 10 days before the observation live session.
Historical PK Count	Total number of PK events in which the streamer participated over the 10 days before the observation live session.
Historical PK Rejections	Number of PK invitations rejected by the streamer over the 10 days before the observation live session.
Historical PK Win Count	Total number of PK events won by the streamer over the 10 days before the observation live session.
Mediators:	
Competition Awareness	Total number of occurrences of retained Competition Awareness keywords during the focal comment window of the observation live session.
Emotional Exclamation	Total number of occurrences of retained Emotional Exclamation keywords during the focal comment window of the observation live session.
Relationship Building	Total number of occurrences of retained Relationship Building keywords during the focal comment window of the observation live session.
Composition Variables:	
Viewer Count	Total number of viewers during the observation live session.
Gifter Count	Total number of viewers who sent at least one gift during the observation live session.
Conversion Rate	Within-session conversion rate, defined as Gifter Count divided by Viewer Count.
Average Gifting per Gifter	Average monetary gifting value per gifting viewer within the observation live session (coded as zero when the session has no gifters).

Notes: All continuous variables are log-transformed, including the dependent variables (*Total Gifting*, *Avg Play Duration per Viewer*, *New Follower Gain*, *Existing Follower Loss*, *Net Follower Gain*, *Viewer-to-viewer Follower Gain*, and *New Loyal Fan Gain*), the continuous control variables (*Tenure Days*, *Follower Count*, *Following Count*, *Loyal Fan Count*, and all Historical measures), the mediators (*Competition Awareness*, *Emotional Exclamation*, and *Relationship Building*), and the composition variables (*Viewer Count*, *Gifter Count*, *Conversion Rate*, and *Average Gifting per Gifter*).

Table 2: Summary Statistics for Study 1: With vs. Without PK

Variable	Obs	Mean	SD	Min	Max
Independent Variables:					
With PK	518,690	0.02	0.14	0.00	1.00
Dependent Variables:					
Total Gifting (Log)	518,690	2.58	3.74	0.00	18.75
Avg Play Duration per Viewer (Log)	518,690	7.62	2.11	0.00	13.48
New Follower Gain (Log)	518,690	2.60	3.22	0.00	15.22
Existing Follower Loss (Log)	518,690	1.80	2.75	0.00	13.75
Net Follower Gain (Log)	518,690	11.99	0.05	0.00	15.10
Viewer-to-viewer Follower Gain (Log)	518,690	0.93	2.13	0.00	12.89
New Loyal Fan Gain (Log)	518,690	0.65	1.78	0.00	12.01
Control Variables:					
Gender	518,690	0.43	0.50	0.00	1.00
Age Category	518,690	5.32	1.19	3.00	7.00

Variable	Obs	Mean	SD	Min	Max
Streamer Content Category	518,690	2.94	0.25	2.00	4.00
Region	518,690	1.21	0.41	0.00	2.00
City Level	518,690	4.55	1.44	0.00	6.00
Device Type	518,690	1.25	0.43	1.00	2.00
App Version	518,690	1.27	1.48	0.00	3.00
Device Price Range	518,690	2.56	0.83	0.00	3.00
Community Type	518,690	2.32	1.29	0.00	4.00
Influencer Account	518,690	0.00	0.03	0.00	1.00
Live Session Category	518,690	6.85	3.64	0.00	13.00
Live Session Start Hour	518,690	13.29	5.85	0.00	23.00
Tenure Days (Log)	518,690	11.90	0.69	4.62	12.85
Follower Count (Log)	518,690	12.51	2.05	0.00	20.37
Following Count (Log)	518,690	10.67	1.93	0.00	13.13
Loyal Fan Count (Log)	518,690	7.44	3.17	0.00	16.43
Historical Total Gifting (Log)	518,690	5.14	4.47	0.00	19.81
Historical Play Duration (Log)	518,690	14.46	3.84	0.00	25.37
Historical View Count (Log)	518,690	11.36	3.14	0.00	22.12
Historical Viewer Count (Log)	518,690	11.08	3.07	0.00	21.79
Historical Like Count (Log)	518,690	9.84	4.27	0.00	21.05
Historical Like Users (Log)	518,690	7.02	2.93	0.00	15.72
Historical Comment Count (Log)	518,690	7.82	3.52	0.00	17.08
Historical Commenters (Log)	518,690	6.77	3.08	0.00	15.37
Historical Voice Comments (Log)	518,690	2.45	3.56	0.00	12.81
Historical Voice Commenters (Log)	518,690	2.06	2.96	0.00	11.01
Historical Share Count (Log)	518,690	2.83	3.58	0.00	16.22
Historical Share Users (Log)	518,690	2.67	3.36	0.00	14.37
Historical PK Count (Log)	518,690	0.31	1.18	0.00	6.55
Historical PK Rejections (Log)	518,690	0.00	0.03	0.00	4.62
Historical PK Win Count (Log)	518,690	0.11	0.70	0.00	6.22
Mediators:					
Competition Awareness (Log)	20,000	0.03	0.16	0.00	2.83
Emotional Exclamation (Log)	20,000	0.12	0.38	0.00	6.08
Relationship Building (Log)	20,000	0.22	0.46	0.00	5.09
Composition Variables:					
Viewer Count (Log)	69,729	9.59	2.81	0.00	18.55
Gifter Count (Log)	69,729	1.65	2.59	0.00	11.83
Conversion Rate (Log)	69,729	0.00	0.02	0.00	0.69
Average Gifting per Gifter (Log)	69,729	0.93	1.93	0.00	17.09

Notes: (a) All variables are measured at the session level in Study 1. (b) To protect data confidentiality, the partner platform scaled the raw values of all continuous variables; categorical variables retain their raw values. (c) The mediation models are estimated on a random subsample of 20,000 of the 26,925 sessions with session-level comment measures, as PROCESS Model 4 with 5,000 bootstrap resamples across seven outcomes is infeasible on the full sample. (d) Viewer-level gifting records are only available from March 1, 2024, onward; composition variables are defined on this 69,729-session window.

3.2.2 Dataset for Study 2: Within Random PK

For Study 2, we retrieve data on livestreaming sessions with Random PK events to examine the relationship between the structure of cooperative social comparison and streamer performance.

Invited PK events and non-PK sessions are outside the Study 2 sampling frame. Our observation window also spans from January 1, 2022, to September 1, 2024. The data are at the livestreaming session level, with each session containing exactly one Random PK event involving a pair of streamers. Coopetitive social comparison in a PK pair is perspective-dependent: designating either streamer as focal reverses the interpretation of the same follower-size gap. For instance, if Streamer A has fewer followers than Streamer B, then A faces a larger opponent while B faces a smaller one. To avoid counting each pair twice, we randomly designate one streamer per pair as the focal streamer.

We analyze PK events from the focal streamer’s perspective, focusing on their outcomes rather than those of their opponents. Our analysis includes only active streamers, defined as those with at least one historical live session during the week preceding the observed session. Further, to avoid mechanical correlation from repeated matchups, we retain only the first PK event for each unique focal-opponent streamer pair. Our final Study 2 dataset comprises 133,191 Random PK livestreaming sessions observed from the focal streamer’s perspective.

We focus on three key independent variables in Study 2. The first variable, *vs. Big Follower Opponent*, indicates whether the opponent’s follower count is greater than or equal to the focal streamer’s follower count. This variable captures whether the focal streamer is at a follower disadvantage, allowing us to examine how this relative standing affects performance. The second variable, *vs. Same Category Opponent*, captures whether the focal and opponent livestreaming sessions belong to the same content category. This variable allows us to examine how category alignment between the two competitors affects the focal streamer’s livestreaming session outcomes. The third variable, *Big Follower Focal (Binary)*, is a binary indicator denoting whether the focal streamer is classified as a ‘Big’ streamer (1 = focal streamer’s follower count is

10,000 or more; 0 = focal streamer's follower count is below 10,000). We use this variable to explore whether the focal streamer's follower size moderates the main effects on the focal streamer's livestreaming outcomes.

The dependent variables in Study 2 include metrics related to streamer performance and audience engagement. Specifically, we incorporate *Focal Win*, a binary indicator of whether the focal streamer won the PK session, to capture the competitive outcome of the event. We also measure the total gifting value and the average value of gifts sent per viewer in a livestreaming session as gifting outcomes. Next, we capture the number of new followers gained, the number of existing followers lost, and the net change in followers, reflecting follower base size fluctuations. Additionally, we examine the number of the opponent's followers gained by the focal streamer and the number of the focal streamer's followers gained by the opponent, as indicators of directional follower attraction between the paired streamers. We include the same categories of control variables as in Study 1, covering streamer characteristics, streaming device usage, session features, prior viewer engagement, historical PK experiences,⁷ and historical revenue.

Further, we include a set of variables used to examine which viewers the gifting response comes from. We partition the viewers of the focal session into three mutually exclusive groups based on which livestream they were watching before the PK began. Both streamers are already broadcasting before the PK matchup in the session; we label as focal viewers those present in the focal streamer's pre-PK live window, and as opponent viewers those present in the opponent streamer's pre-PK live window. All remaining viewers of the focal session who joined only after the PK started and were not present in either streamer's pre-PK audience are labeled as other

⁷ *Historical PK Rejections* is not included in Study 2 because rejections occur only in Invited PK events; Random PK events, by design, do not involve invitations and therefore cannot be rejected.

viewers. For each group, we measure the gifts the focal streamer receives from the start of the PK to the end of the focal session, that is, over the PK window and the remainder of the session combined (*Focal Viewer Gifting*, *Opponent Viewer Gifting*, and *Other Viewer Gifting*).

Table 3 presents the list of variables and their descriptions for Study 2, and Table 4 shows the descriptive statistics of these variables. As in Study 1, unless otherwise noted, all continuous dependent, control, and composition variables are log-transformed to reduce distributional skewness, while categorical variables are encoded using dummy indicators. Percentages reported in the results section are calculated as $\exp(\beta) - 1$ for log-transformed outcomes.

Table 3: Variable Descriptions for Study 2: Within Random PK

Variables	Descriptions
Independent Variables:	
vs. Big Follower Opponent	Indicates whether the opponent's follower count is greater than or equal to the focal streamer's follower count (1 = yes; 0 = no).
vs. Same Category Opponent	Indicates whether the focal and opponent livestreaming sessions belong to the same content category (0 = no; 1 = yes).
Big Follower Focal (Binary)	Indicates whether the focal streamer has a large follower base (1 = follower count $\geq 10,000$; 0 = follower count $< 10,000$).
Dependent Variables:	
Focal Win	Indicates whether the focal streamer won the PK session (1 = win; 0 = loss). Draw events are excluded when this variable is used in analysis.
Total Gifting	Total monetary value of gifts received during the focal session.
Average Gifting per Viewer	Average monetary value of gifts sent per viewer during the focal session.
Opponent Follower Attraction	Number of followers who originally followed the opponent but started following the focal streamer during the focal session.
Focal Follower Attraction	Number of followers who originally followed the focal streamer but started following the opponent during the focal session.
New Follower Gain	Number of new followers gained by the focal streamer during the focal session.
Existing Follower Loss	Number of existing followers lost by the focal streamer during the focal session.
Net Follower Gain	Net change in the number of followers for the focal streamer during the focal session. Because the net change can be negative, it is shifted by its sample minimum before the log transformation, $\log(y - y_{\min} + 1)$.
Control Variables:	
Gender	Gender of the streamer (0 = Male; 1 = Female).
Age Category	Age category of the streamer (3 = 18–23; 4 = 24–30; 5 = 31–40; 6 = 41–49; 7 = 50 and above).
Streamer Content Category	Content category of the streamer (1 = Expert; 2 = E-commerce; 3 = Show; 4 = Game).

Variables	Descriptions
Region	Region of the streamer (0 = Unknown; 1 = Northern; 2 = Southern).
City Level	Tier of the streamer's city (0 = Unknown; 1 = First-tier; 2 = New first-tier; 3 = Second-tier; 4 = Third-tier; 5 = Fourth-tier; 6 = Fifth-tier).
Device Type	Type of device used (1 = Android; 2 = iOS).
App Version	App version used (0 = Unknown; 1 = Lite; 2 = Express; 3 = Regular).
Device Price Range	Device price range (0 = Budget; 1 = Mid-range; 2 = High-end; 3 = Premium).
Community Type	Community classification (0 = Unknown; 1 = Urban; 2 = Foreign; 3 = Rural; 4 = Township).
Influencer Account	Indicates if the streamer is a verified influencer (0 = No; 1 = Yes).
Live Session Category	Content category of the focal session, coded 1 to 13; focal sessions whose category is unknown (code 0) are excluded from the Study 2 dataset.
Live Session Start Hour	Starting hour of the focal session.
Tenure Days	Number of days since registration (streamer tenure).
Follower Count	Total follower count before the focal session.
Following Count	Number of users followed by the streamer before the focal session.
Loyal Fan Count	Count of loyal fans before the focal session.
Historical Total Gifting	Total gift value in prior Random PK sessions over the 10 days before the focal session.
Historical Play Duration	Total play duration in prior Random PK sessions over the 10 days before the focal session.
Historical View Count	Total number of views in prior Random PK sessions over the 10 days before the focal session.
Historical Viewer Count	Total viewer count in prior Random PK sessions over the 10 days before the focal session.
Historical Like Count	Total like count in prior Random PK sessions over the 10 days before the focal session.
Historical Like Users	Number of unique users who liked content in prior Random PK sessions over the 10 days before the focal session.
Historical Comment Count	Total comment count in prior Random PK sessions over the 10 days before the focal session.
Historical Commenters	Number of unique commenters in prior Random PK sessions over the 10 days before the focal session.
Historical Voice Comments	Total voice comment count in prior Random PK sessions over the 10 days before the focal session.
Historical Voice Commenters	Number of unique voice commenters in prior Random PK sessions over the 10 days before the focal session.
Historical Share Count	Total share count in prior Random PK sessions over the 10 days before the focal session.
Historical Share Users	Number of unique users who shared content in prior Random PK sessions over the 10 days before the focal session.
Historical PK Count	Total number of PK events in which the streamer participated over the 10 days before the focal session.
Historical PK Win Count	Total number of PK events won by the streamer over the 10 days before the focal session.

Variables	Descriptions
Composition Variables:	
Focal Viewer Gifting	Total monetary value of gifts received during and after the PK window of the focal session (excluding the pre-PK portion) from viewers who come from the focal streamer's channel.
Opponent Viewer Gifting	Total monetary value of gifts received during and after the PK window of the focal session (excluding the pre-PK portion) from viewers who come from the opponent streamer's channel.
Other Viewer Gifting	Total monetary value of gifts received during and after the PK window of the focal session (excluding the pre-PK portion) from viewers who come from neither the focal nor the opponent streamer's channel.

Notes: (a) All continuous variables are log-transformed, including the dependent variables (*Total Gifting*, *Average Gifting per Viewer*, *Opponent Follower Attraction*, *Focal Follower Attraction*, *New Follower Gain*, *Existing Follower Loss*, and *Net Follower Gain*), the continuous control variables (*Tenure Days*, *Follower Count*, *Following Count*, *Loyal Fan Count*, and all Historical measures), and the composition variables (*Focal Viewer Gifting*, *Opponent Viewer Gifting*, and *Other Viewer Gifting*). *vs. Big* = *vs. Big Follower Opponent*; *vs. Same* = *vs. Same Category Opponent*. (b) Because Study 2 focuses exclusively on Random PK events, the Historical measures listed in this table are constructed from prior Random PK sessions observed in the Study 2 dataset during the preceding 10 days; Invited PK and non-PK sessions are not included.

Table 4: Summary Statistics for Study 2: Within Random PK

Variable	Obs	Mean	SD	Min	Max
Independent Variables:					
vs. Big Follower Opponent	133,191	0.49	0.50	0.00	1.00
vs. Same Category Opponent	133,191	0.33	0.47	0.00	1.00
Big Follower Focal (Binary)	133,191	0.18	0.39	0.00	1.00
Dependent Variables:					
Focal Win	61,495	0.49	0.50	0.00	1.00
Total Gifting (Log)	133,191	3.08	4.20	0.00	16.57
Average Gifting per Viewer (Log)	133,191	1.06	1.81	0.00	11.45
Opponent Follower Attraction (Log)	133,191	0.05	0.49	0.00	10.13
Focal Follower Attraction (Log)	133,191	0.05	0.50	0.00	9.43
New Follower Gain (Log)	133,191	1.11	2.28	0.00	12.72
Existing Follower Loss (Log)	133,191	0.92	2.05	0.00	12.26
Net Follower Gain (Log)	133,191	11.01	0.04	0.00	12.57
Control Variables:					
Gender	133,191	0.46	0.50	0.00	1.00
Age Category	133,191	5.28	1.34	3.00	7.00
Streamer Content Category	133,191	2.99	0.17	1.00	4.00
Region	133,191	1.26	0.44	0.00	2.00
City Level	133,191	4.45	1.49	0.00	6.00
Device Type	133,191	1.30	0.46	1.00	2.00
App Version	133,191	1.11	1.45	0.00	3.00
Device Price Range	133,191	2.54	0.84	0.00	3.00
Community Type	133,191	2.31	1.27	0.00	4.00
Influencer Account	133,191	0.00	0.01	0.00	1.00
Live Session Category	133,191	8.55	2.94	1.00	13.00
Live Session Start Hour	133,191	15.01	5.88	0.00	23.00
Tenure Days (Log)	133,191	11.79	0.82	4.62	12.97
Follower Count (Log)	133,191	12.27	1.97	0.00	20.77
Following Count (Log)	133,191	10.69	2.02	0.00	13.13
Loyal Fan Count (Log)	133,191	6.95	3.42	0.00	17.08
Historical Total Gifting (Log)	133,191	0.12	1.02	0.00	15.21
Historical Play Duration (Log)	133,191	0.54	2.49	0.00	20.52
Historical View Count (Log)	133,191	0.43	1.99	0.00	16.56

Variable	Obs	Mean	SD	Min	Max
Historical Viewer Count (Log)	133,191	0.42	1.95	0.00	16.13
Historical Like Count (Log)	133,191	0.26	1.47	0.00	16.27
Historical Like Users (Log)	133,191	0.19	1.06	0.00	10.79
Historical Comment Count (Log)	133,191	0.20	1.18	0.00	12.52
Historical Commenters (Log)	133,191	0.18	1.03	0.00	11.47
Historical Voice Comments (Log)	133,191	0.02	0.31	0.00	9.38
Historical Voice Commenters (Log)	133,191	0.01	0.26	0.00	7.65
Historical Share Count (Log)	133,191	0.02	0.31	0.00	10.42
Historical Share Users (Log)	133,191	0.02	0.29	0.00	9.63
Historical PK Count (Log)	133,191	0.23	1.03	0.00	7.00
Historical PK Win Count (Log)	133,191	0.06	0.52	0.00	5.99
Composition Variables:					
Focal Viewer Gifting (Log)	27,510	1.24	3.29	0.00	16.90
Opponent Viewer Gifting (Log)	27,510	0.02	0.39	0.00	12.08
Other Viewer Gifting (Log)	27,510	7.88	3.67	0.00	20.92

Notes: (a) All variables are measured at the session level in Study 2. (b) To protect data confidentiality, the partner platform scaled the raw values of all continuous variables; categorical variables retain their raw values. (c) *Focal Win* excludes draw events; its observation count reflects a restricted subsample (N = 61,495). (d) Viewer-level gifting records are only available from March 1, 2024, onward; composition variables are defined on this 27,510-session window. (e) *Opponent Follower Attraction (Log)* counts the opponent's followers who start following the focal streamer; *Focal Follower Attraction (Log)* counts the focal streamer's followers who start following the opponent. (f) The minimum value of *Streamer Content Category* differs across studies (1 in Study 2, 2 in Study 1) because Expert-category streamers (category = 1) are absent from the Study 1 sample due to its 1% random sampling. *vs. Big = vs. Big Follower Opponent*; *vs. Same = vs. Same Category Opponent*. (g) *Total Gifting* is a platform-provided measure for the entire livestream. The three viewer-source measures are reconstructed from viewer-level gifting records during and after the PK for the 27,510-session subsample and are on a separate scale; their levels are therefore not directly comparable with, or intended to sum to, *Total Gifting*.

3.3 Empirical Strategy

3.3.1 Study 1: With vs. Without PK

Study 1 first estimates the average effect of PK participation on streamer performance. We then decompose the gifting effect to see whether it comes from more viewers, from a larger share of viewers who gift, or from larger gifts per gifter, and we use a mediation analysis of viewer comments to ask whether the PK effect is accompanied by comments that refer to the contest rather than by comments that express excitement or social bonding. The empirical analyses in Study 1 contain the following steps.

Main Effects. In Study 1, we seek to estimate the impact of participating in PK events on streamers' performance. In doing so, it is important to address potential endogeneity concerns

that may arise from omitted variable bias. For example, streamer characteristics, prior engagement and PK history, and session content and timing could influence both a streamer’s likelihood of participating in PK events and the resulting outcomes. We therefore employ Double Machine Learning (DML), which uses machine-learning algorithms to estimate the relationship between the treatment variable and the outcome while flexibly adjusting for these observed confounders (Chernozhukov et al., 2018; Ye et al., 2025). DML suits this setting for three reasons: the categorical controls expand into a high-dimensional set of indicator variables once encoded; the relationships between the covariates and both PK participation and session performance may be nonlinear; and its orthogonalization and cross-fitting procedures reduce first-order sensitivity of the treatment-effect estimate to errors in the machine-learning estimation of the nuisance functions. Specifically, we employ partially linear regression to estimate the effect of PK participation on different outcomes while controlling for high-dimensional covariates, as specified in Equations (1) and (2):

$$Y = D\theta_0 + g_0(X) + \zeta, \quad E[\zeta \mid D, X] = 0, \quad (1)$$

$$D = m_0(X) + V, \quad E[V \mid X] = 0, \quad (2)$$

where Y is the outcome variable of interest; D is a binary indicator for PK event participation; θ_0 , the parameter of interest, is the effect of PK participation on the outcome; and X is a set of high-dimensional control variables.⁸ The function $g_0(X)$ captures potential nonlinear effects of the control variables, and the function $m_0(X)$ represents the conditional expectation of treatment assignment given the covariates, while ζ and V represent stochastic error terms.

⁸ Control variables are selected based on contextual understanding and consultations with the partner platform, as explained in Section 3.2.1, to account for key streamer-specific and historical factors influencing participation, engagement, and gifting outcomes. We include as many relevant controls as possible, based on both theoretical considerations and empirical feasibility.

To estimate θ_0 , we apply the DML procedure, which consists of three steps. In the first step, we estimate the conditional expectations of Y and D given X , known as the nuisance functions: $l(X) = E[Y | X]$, $m_0(X) = E[D | X]$. These functions are estimated using machine learning methods to flexibly capture nonlinear relationships. In the second step, residualization (orthogonalization), we construct residualized versions of Y and D : $\tilde{Y} = Y - \hat{l}(X)$, $\tilde{D} = D - \hat{m}_0(X)$. This orthogonalization step removes the part of Y and D explained by X , so that θ_0 is estimated from the variation in D that the controls do not explain. In the third step, to estimate the treatment effect, we substitute these residualized variables to obtain $\tilde{Y} = \theta_0 \tilde{D} + \zeta$, where $E[\zeta \tilde{D}] = 0$. Under this orthogonality condition, we estimate θ_0 using ordinary least squares (OLS): $\hat{\theta}_0 = (\sum_i \tilde{D}_i^2)^{-1} \sum_i \tilde{D}_i \tilde{Y}_i$.

To estimate the nuisance functions $\hat{l}(X)$ and $\hat{m}_0(X)$, we use five different machine learning models: Least absolute shrinkage and selection operator (LASSO), random forests, regression trees, boosted trees, and neural networks, which allow us to flexibly capture the relationships between the control variables and both the outcome and treatment assignment. We use 5-fold cross-fitting with folds grouped by streamer, so that all of a streamer's sessions fall in the same fold. Cross-fitting removes the bias that plug-in nuisance estimates would otherwise introduce into the treatment-effect estimator and, together with the orthogonalization above, enables root-n consistent and asymptotically normal inference. Standard errors are clustered at the streamer level. In addition, Online Appendix C reports robustness checks of the main estimates: two-way fixed-effects regressions; propensity score matching (PSM) using a generalized linear model estimator with varying nearest-neighbor ratios (1:1, 1:2, and 1:10) and coarsened exact matching (CEM), each followed by a two-way fixed-effects regression on the

matched sample, with balance diagnostics; a matched pre-PK engagement analysis; and a comparison of streamers who accepted and declined PK invitations.

Gifting Response Decomposition. The cooperative and competitive dimensions of PK may manifest in different margins of the gifting response. The joint split-screen performance exposes each streamer to a broader audience, so an increase in viewer count would be consistent with the cooperative dimension of PK. At the same time, because viewers' gifts determine the contest outcome, an increase in gifting among those who already give would be consistent with its competitive dimension. A herding explanation suggests a different pattern: observing others gift may encourage additional viewers to begin gifting, which would be reflected in a higher gifting conversion rate.

To distinguish these patterns, we use the identity $\log(\text{Total Gifting}) = \log(\text{Viewer Count}) + \log(\text{Conversion Rate}) + \log(\text{Average Gifting per Gifter})$ and estimate the main effect with the same DML and 5-fold cross-fitting on four decomposition outcomes:⁹ *Viewer Count*, which measures how many people watch (the extensive margin); *Gifter Count*, which measures the size of the contributing base; *Conversion Rate* (defined as *Gifter Count* divided by *Viewer Count*), which measures the share of viewers who gift; and *Average Gifting per Gifter*, which measures how much each gifter spends (the intensive margin).¹⁰ The component estimates show which of these margins move alongside the gifting gain and whether the pattern is consistent with the cooperative and competitive dimensions of PK or with herding.

⁹ In levels, $\text{Total Gifting} = \text{Viewer Count} \times (\text{Gifter Count}/\text{Viewer Count}) \times (\text{Total Gifting}/\text{Gifter Count})$ for sessions with at least one gifter. Each component is estimated as a separate outcome under the $\log(1 + y)$ transformation used throughout, and many sessions have no gifters, so the estimated component effects need not sum to the estimated effect on *Total Gifting*; we read them as evidence of which margins respond.

¹⁰ Because viewer-level gifting records are available only in a restricted window of the partner platform's data archive, this analysis is conducted on a 69,729-session subsample.

Discursive Engagement Mediation. The decomposition identifies which margins move with the gifting gain but does not indicate what viewers are responding to. We therefore examine viewer comments to assess whether the observed discourse is more consistent with the competitive aspect of cooperative social comparison than with general excitement or social bonding. We collect comments during a 5-minute focal window for each session (the actual PK window for treated sessions and a pseudo-PK window matched to the typical timing of PK events for control sessions).¹¹ Following a dictionary-based content-analysis approach, we construct three measures of viewer discourse. *Competition Awareness* captures the occurrence of keywords referring to the contest format, the opponent, and contest outcomes. *Emotional Exclamation* captures the occurrence of keywords expressing excitement or other spontaneous affect. *Relationship Building* captures the occurrence of keywords referring to the streamer or other viewers as friends, family, or fan-group members. The keyword dictionaries are developed through LLM-assisted categorization of candidate keywords using representative comment contexts, followed by manual review and refinement. Only keywords judged to be clear indicators of one of the three theoretically defined categories are retained. For each session, each mediator is calculated as $\log(1 + \text{the number of occurrences of the corresponding keywords within the focal window})$. The complete keyword dictionaries and construction procedure are reported in Online Appendix D. Using the three measures as parallel mediators, we estimate the indirect effect of PK participation through each category with PROCESS Model 4 (Hayes, 2022). Indirect effects are estimated via 5,000 percentile bootstrap replications on a random subsample of 20,000 of the 26,925 sessions with session-level comment measures, due to

¹¹ The pseudo-PK window for a non-PK session is defined as follows: across all treated sessions, we compute the ratio of PK start time to total session duration and take the median of these ratios; this median ratio is then applied to each non-PK session to determine the start of its pseudo-PK 5-minute window.

computational constraints, and an indirect effect is treated as statistically significant when its 95% bootstrap confidence interval excludes zero.

3.3.2 Study 2: Within Random PK

Study 2 examines the effects of two dimensions of the structure of coopetitive social comparison on focal-streamer outcomes within Random PK events. The gifting source decomposition in Study 2 then asks which viewers the additional gifting comes from. We also conduct a heterogeneity analysis to examine whether the focal streamer's follower size moderates the main effects of the two comparison structures on the focal streamer's outcomes. We analyze the data in Study 2 following the steps below.

Main Effects. First, we leverage the platform-generated pairing of streamers in Random PK events and focus on two dimensions of the structure of coopetitive social comparison: follower-size comparison and content-category alignment. Before proceeding, we compare focal and opponent streamers on the observable characteristics available to us. The results, reported in Online Appendix H, show absolute standardized mean differences below 0.1 for most observables. We then employ OLS regressions to examine the effects of the two structures on the focal streamer's performance, following Equations (3) and (4):

$$Y_i = \beta_1 \cdot vs_Big_Follower_Opponent_i + \gamma \cdot X_i + \delta_{date} + \varepsilon_i \quad (3)$$

$$Y_i = \beta_1 \cdot vs_Same_Category_Opponent_i + \gamma \cdot X_i + \delta_{date} + \varepsilon_i \quad (4)$$

where Y_i represents an outcome variable of interest at the livestreaming-session level. The independent variable $vs_Big_Follower_Opponent_i$ indicates whether the opponent streamer's follower size is greater than or equal to the focal streamer's follower size (1 = yes; 0 = no), and $vs_Same_Category_Opponent_i$ indicates whether the focal and opponent livestreaming sessions belong to the same content category (1 = same; 0 = different). The vector X_i includes the

set of control variables, with coefficient vector γ ; δ_{date} denotes date fixed effects; β_1 is the coefficient of interest; and ε_i is the error term. We cluster standard errors at the streamer level to account for within-streamer correlation in the error terms.

Gifting Source Decomposition. To examine how gifting from each viewer segment responds to comparison structure, we re-estimate Equations (3) and (4) on the three viewer-source measures defined in Section 3.2.2, *Focal Viewer Gifting*, *Opponent Viewer Gifting*, and *Other Viewer Gifting*, each measured from the start of the PK to the end of the focal session and log-transformed. An increase in gifting from the focal streamer’s own audience would be consistent with the competitive dimension of PK, as these are the viewers who were in the focal streamer’s room when the contest began. The opponent’s viewers and new arrivals are the two audiences through which the cooperative dimension can appear: the jointly produced performance exposes each streamer to the other’s audience and draws in viewers who were in neither room. This decomposition is conducted on the subsample of 27,510 sessions for which viewer-level gifting records are available.

Heterogeneity by Streamer Characteristics. Finally, we examine the moderating effect of the focal streamer’s follower base size, estimating Equations (5) and (6):

$$Y_i = \beta_1 \cdot vs.Big_i + \beta_2 \cdot Big_Focal_i + \beta_3 \cdot (vs.Big_i \times Big_Focal_i) + \gamma \cdot X_i + \delta_{date} + \varepsilon_i \quad (5)$$

$$Y_i = \beta_1 \cdot vs.Same_i + \beta_2 \cdot Big_Focal_i + \beta_3 \cdot (vs.Same_i \times Big_Focal_i) + \gamma \cdot X_i + \delta_{date} + \varepsilon_i \quad (6)$$

Here, $vs.Big_i$ and $vs.Same_i$ abbreviate $vs.Big_Follower_Opponent_i$ and $vs.Same_Category_Opponent_i$ from Equations (3) and (4), and Big_Focal_i (*Big Follower Focal (Binary)*) indicates whether the focal streamer is classified as a ‘Big’ streamer; β_2 and β_3

are the coefficients on the focal-size indicator and on its interaction with the comparison structure. The vector X_i includes the set of control variables, and δ_{date} denotes date fixed effects. Online Appendix I reports two robustness checks: a continuous measure of the focal streamer’s follower size as an alternative moderator, and a continuous follower-difference measure in place of the binary follower-size comparison.

4. RESULTS

4.1 Study 1: With vs. Without PK

4.1.1 Main Effects

In Study 1, we estimate Equation (1) to examine the impact of participating in PK events on streamer performance. Table 5 presents the results of the DML estimates. Across all five learners, total gifting increases by at least 328.9% ($\hat{\beta} = 1.456$, $p < 0.001$), average play duration by 106.3% ($\hat{\beta} = 0.724$, $p < 0.001$), new follower acquisition by 131.9% ($\hat{\beta} = 0.841$, $p < 0.001$), viewer-to-viewer interactions by 102.4% ($\hat{\beta} = 0.705$, $p < 0.001$), and loyal fan acquisition by 22.5% ($\hat{\beta} = 0.203$, $p < 0.001$). Although PK events also elevate existing follower loss, net follower growth remains positive under every learner (smallest estimate $\hat{\beta} = 0.00204$, $p < 0.01$). These results show that PK participation substantially improves gifting, viewer engagement, and follower acquisition.

Table 5: The Effects of PK Event Participation on Streamer Performance

	Total Gifting (Log)	Avg Play Duration per Viewer (Log)	New Follower Gain (Log)	Viewer-to- viewer Follower Gain (Log)	New Loyal Fan Gain (Log)	Existing Follower Loss (Log)	Net Follower Gain (Log)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
LASSO	1.612*** (0.062)	0.724*** (0.020)	0.959*** (0.043)	0.768*** (0.050)	0.245*** (0.030)	0.608*** (0.038)	0.00264*** (0.00076)
Random Forest	1.696*** (0.080)	0.725*** (0.022)	0.922*** (0.043)	0.791*** (0.054)	0.238*** (0.030)	0.661*** (0.043)	0.00226*** (0.00062)
Regression Trees	1.456*** (0.064)	0.732*** (0.024)	0.841*** (0.045)	0.852*** (0.055)	0.203*** (0.031)	0.582*** (0.040)	0.00204** (0.00068)

	Total Gifting (Log)	Avg Play Duration per Viewer (Log)	New Follower Gain (Log)	Viewer-to- viewer Follower Gain (Log)	New Loyal Fan Gain (Log)	Existing Follower Loss (Log)	Net Follower Gain (Log)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Boosted Trees	1.470*** (0.055)	0.728*** (0.018)	0.989*** (0.038)	0.708*** (0.042)	0.252*** (0.025)	0.625*** (0.034)	0.00313*** (0.00074)
Neural Network	1.485*** (0.054)	0.731*** (0.019)	0.985*** (0.038)	0.705*** (0.042)	0.247*** (0.025)	0.613*** (0.033)	0.00253** (0.00089)
Obs	518,690	518,690	518,690	518,690	518,690	518,690	518,690

Notes: Cross-fitting folds are grouped by streamer; streamer-clustered standard errors in parentheses; + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

4.1.2 Gifting Response Decomposition

We next examine which margins move alongside the gifting gain: the number of viewers, the share of viewers who gift, or the size of gifts per gifter. As set out in Section 3.3.1, an increase in viewer count would be consistent with the cooperative dimension of PK, an increase in gifts per gifter with its competitive dimension, and a higher gifting conversion rate with herding.

Table 6 reports the decomposition results. Across all five learners, *Viewer Count* rises by 147.4–170.2% ($\hat{\beta} = 0.906$ to 0.994) and *Average Gifting per Gifter* rises by 36.6–51.7% ($\hat{\beta} = 0.312$ to 0.417), both highly significant. By contrast, the estimated change in *Conversion Rate* is economically negligible and not statistically significant. Two margins therefore rise alongside the gifting gain: more people watch, and those who gift give more.¹² The first is consistent with the cooperative dimension of PK, since the split-screen performance exposes each streamer to the other’s audience; the second is consistent with its competitive dimension, since a supporter’s gifts determine whether the preferred streamer wins. The share of viewers who gift shows no statistically detectable change, providing no evidence for a herding pattern operating through the conversion of viewers into gifters.

¹² Table 6 is estimated on the 69,729-session window for which viewer-level gifting records exist (Section 3.3.1); the estimated effect on *Total Gifting* in this window (0.698 to 0.876) is smaller than in the full sample of Table 5.

Table 6: Decomposition of the Gifting Response (Study 1)

	Total Gifting (Log)	Viewer Count (Log)	Gifter Count (Log)	Conversion Rate (Log)	Average Gifting per Gifter (Log)
	(1)	(2)	(3)	(4)	(5)
LASSO	0.796*** (0.157)	0.994*** (0.058)	0.437*** (0.089)	-0.00003 (0.00060)	0.365*** (0.076)
Random Forest	0.876*** (0.174)	0.987*** (0.059)	0.466*** (0.096)	0.00010 (0.00058)	0.408*** (0.084)
Regression Trees	0.797*** (0.158)	0.967*** (0.060)	0.387*** (0.092)	-0.00018 (0.00076)	0.417*** (0.077)
Boosted Trees	0.704*** (0.153)	0.987*** (0.055)	0.400*** (0.087)	-0.00001 (0.00060)	0.312*** (0.074)
Neural Network	0.698*** (0.154)	0.906*** (0.053)	0.420*** (0.083)	0.00019 (0.00054)	0.332*** (0.073)
Obs	69,729	69,729	69,729	69,729	69,729

Notes: (a) To make the decomposition comparable across margins using a common sample, *Average Gifting per Gifter* is coded as $\log(1 + \text{gift amount/gifter count})$ when *Gifter Count* > 0 and as zero otherwise, and all 69,729 sessions are retained. (b) Cross-fitting folds are grouped by streamer; streamer-clustered standard errors in parentheses; + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

4.1.3 Discursive Engagement Mediation

We next examine whether what viewers write during the focal window is more consistent with the competitive aspect of coepetitive social comparison than with general excitement or social bonding. As described in Section 3.3.1, we construct three session-level measures based on the occurrence of category-specific keywords: *Competition Awareness*, *Emotional Exclamation*, and *Relationship Building*.

Table 7 reports the results. The indirect effect through *Competition Awareness* is positive, with a confidence interval that excludes zero, for all seven outcomes, and it is the largest of the three indirect effects in every case. The indirect effects through *Emotional Exclamation* and *Relationship Building* have confidence intervals that include zero for every outcome. The total indirect effect is significant for five of the seven outcomes (total gifting, average play duration per viewer, viewer-to-viewer follower gain, new loyal fan gain, and net follower gain); its confidence interval includes zero for new follower gain and existing follower loss. The direct effects remain large and significant.

The indirect association therefore runs through expressions about the contest rather than through expressions of excitement or of connection with the streamer. This pattern is consistent with the competitive aspect of co-competitive social comparison rather than with general excitement or social bonding. Together with the decomposition, Study 1 shows that viewer count and gifts per gifter rise with PK participation and that viewers’ comments during PK reflect awareness of the contest.

Table 7: Discursive Engagement Mediation Analysis (Study 1)

	Total Gifting (Log)	Avg Play Duration per Viewer (Log)	New Follower Gain (Log)	Viewer-to- viewer Follower Gain (Log)	New Loyal Fan Gain (Log)	Existing Follower Loss (Log)	Net Follower Gain (Log)
Mediation (Total Indirect Effect)	0.066	0.014	0.029	0.070	0.045	0.026	0.001
	[0.013, 0.121]	[0.007, 0.022]	[-0.006, 0.066]	[0.030, 0.113]	[0.016, 0.078]	[-0.002, 0.055]	[0.001, 0.003]
– Competition Awareness	0.065	0.013	0.034	0.072	0.048	0.030	0.001
	[0.035, 0.102]	[0.007, 0.020]	[0.014, 0.059]	[0.042, 0.106]	[0.025, 0.074]	[0.012, 0.052]	[0.001, 0.002]
– Emotional Exclamation	0.009	0.001	0.003	0.004	0.002	0.002	0.0001
	[-0.013, 0.034]	[-0.002, 0.004]	[-0.004, 0.011]	[-0.005, 0.014]	[-0.003, 0.009]	[-0.002, 0.006]	[-0.0001, 0.0003]
– Relationship Building	-0.009	0.0003	-0.008	-0.006	-0.005	-0.006	-0.0001
	[-0.035, 0.018]	[-0.001, 0.001]	[-0.032, 0.017]	[-0.022, 0.011]	[-0.019, 0.010]	[-0.022, 0.011]	[-0.0003, 0.0001]
Direct Effect	1.253	0.139	0.797	0.944	0.337	0.519	0.007
	[0.877, 1.628]	[0.075, 0.202]	[0.538, 1.056]	[0.684, 1.204]	[0.127, 0.547]	[0.282, 0.756]	[0.001, 0.012]
Total Effect	1.318	0.153	0.827	1.014	0.382	0.545	0.008
	[0.941, 1.695]	[0.090, 0.215]	[0.569, 1.084]	[0.752, 1.275]	[0.172, 0.592]	[0.310, 0.780]	[0.002, 0.014]

Notes: (a) Independent variable = *With PK*. Mediators = *Competition Awareness (Log)*, *Emotional Exclamation (Log)*, and *Relationship Building (Log)*. Dependent variables are the seven streamer-performance outcomes: *Total Gifting (Log)*, *Avg Play Duration per Viewer (Log)*, *New Follower Gain (Log)*, *Viewer-to-viewer Follower Gain (Log)*, *New Loyal Fan Gain (Log)*, *Existing Follower Loss (Log)*, and *Net Follower Gain (Log)*. (b) Mediation analyses are conducted using PROCESS Model 4 (Hayes, 2022) on a random subsample of 20,000 of the 26,925 sessions for which session-level comment measures are available. (c) Reported effects are the total indirect, mediator-specific indirect, direct, and total effects. (d) Square brackets report 95% percentile bootstrap confidence intervals from 5,000 bootstrap resamples. An indirect effect is statistically significant at the 5% level when its interval excludes zero.

4.1.4 Additional Analysis

We conduct two additional analyses to examine the scope of the Study 1 findings. First, we estimate the effects of Invited and Random PK in one specification relative to sessions without

PK. Both formats show positive point estimates on the main outcomes, indicating that the main findings are not attributable to a single PK format; the Invited PK coefficients are larger for total gifting and for the five follower and fan-group outcomes, with no statistically detectable difference for play duration (Online Appendix F). Second, we examine additional outcomes beyond the PK itself. PK participation increases the likelihood of participating in another PK within 30 days and of changing the content category and starting hour of the next session, and it increases the number of previously inactive viewers who return during the session and the number of views over the following 30 days; however, we find no consistent evidence of increased overall streamer retention or loyal-fan growth (Online Appendix G).

In summary, Study 1 establishes three findings. First, PK participation generates large gains across multiple performance outcomes, with the smallest estimates across the five DML learners implying increases of 328.9% in total gifting, 106.3% in average play duration per viewer, and 131.9% in new follower gain. Second, viewer count and average gifting per gifter rise alongside the gifting gain, while the share of viewers who gift shows no statistically detectable change; the two rising margins are consistent with the cooperative and competitive dimensions of PK, and the absence of a statistically detectable change in the conversion rate provides no evidence for a herding pattern operating through the conversion of viewers into gifters. Third, a mediation analysis of viewer comments shows a significant indirect association through competition awareness, reflected in comments that refer to the contest, whereas comments expressing excitement or affiliation show none, which is consistent with the competitive aspect of cooperative social comparison rather than with general excitement or social bonding. A behavioral-engagement mediation analysis, in which views, likes, comments, and shares carry a large share of the PK effect, is reported in Online Appendix E.

4.2 Study 2: Within Random PK

4.2.1 Main Effects

In Study 2, we investigate how the structure of cooperative social comparison within Random PK events influences focal streamers' outcomes. We begin by estimating Equation (3) to examine the impact of follower-size comparison; Table 8 reports the results. Facing a bigger opponent significantly decreases the focal streamer's win probability by 5.6 percentage points but increases total gifting by 59.20% ($\hat{\beta} = 0.465$, $p < 0.001$) and average gifting per viewer by 19.12% ($\hat{\beta} = 0.175$, $p < 0.001$). It also significantly increases cross-streamer follower exchange and the focal streamer's own follower gains and losses, yielding a positive net effect on the follower base. The result that facing a bigger opponent reduces the focal streamer's probability of winning while increasing total gifting and the follower base illustrates the cooperative nature of PK outcomes. Thus, a lower probability of winning can coexist with gains in the financial and audience acquisition dimensions.

Table 8: The Impact of Follower-Size Comparison on Streamer Performance

Panel A: Gifting Outcomes					
	Focal Win	Total Gifting (Log)	Average Gifting per Viewer (Log)		
	(1)	(2)	(3)		
vs. Big	-0.056*** (0.004)	0.465*** (0.021)	0.175*** (0.010)		
Obs	61,495	133,191	133,191		
Controls	Yes	Yes	Yes		
Date FE	Yes	Yes	Yes		
Panel B: Follower Dynamics					
	Opponent Follower Attraction (Log)	Focal Follower Attraction (Log)	New Follower Gain (Log)	Existing Follower Loss (Log)	Net Follower Gain (Log)
	(1)	(2)	(3)	(4)	(5)
vs. Big	0.051*** (0.003)	0.022*** (0.003)	0.141*** (0.012)	0.103*** (0.011)	0.001*** (0.0001)
Obs	133,191	133,191	133,191	133,191	133,191
Controls	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors clustered at the streamer level in parentheses; + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. vs. Big = vs. Big Follower Opponent. Focal Win excludes draw events, hence N = 61,495.

Next, we turn to Equation (4) to evaluate the relationship between content-category alignment and focal streamers' performance. Table 9 shows that facing a same-category opponent does not significantly affect win probability but increases total gifting by 18.29% ($\hat{\beta} = 0.168$, $p < 0.001$) and average gifting per viewer by 10.08% ($\hat{\beta} = 0.096$, $p < 0.001$). It also increases cross-streamer follower exchange, while proportionally reducing both follower gains and losses, with a small, marginally significant decrease in net follower gain.

Table 9: The Impact of Content-Category Alignment on Streamer Performance

Panel A: Gifting Outcomes					
	Focal Win	Total Gifting (Log)		Average Gifting per Viewer (Log)	
	(1)	(2)		(3)	
vs. Same	0.008 (0.005)	0.168*** (0.022)		0.096*** (0.010)	
Obs	61,495	133,191		133,191	
Controls	Yes	Yes		Yes	
Date FE	Yes	Yes		Yes	
Panel B: Follower Dynamics					
	Opponent Follower Attraction (Log)	Focal Follower Attraction (Log)	New Follower Gain (Log)	Existing Follower Loss (Log)	Net Follower Gain (Log)
	(1)	(2)	(3)	(4)	(5)
vs. Same	0.020*** (0.003)	0.019*** (0.003)	-0.045*** (0.013)	-0.045*** (0.011)	-0.00034+ (0.00020)
Obs	133,191	133,191	133,191	133,191	133,191
Controls	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors clustered at the streamer level in parentheses; + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. vs. Same = vs. Same Category Opponent. Focal Win excludes draw events, hence N = 61,495.

Taken together, Tables 8 and 9 show that PK outcomes can diverge across dimensions. Facing a bigger opponent reduces the focal streamer's probability of winning but increases gifting and net follower growth, whereas facing a same-category opponent increases gifting and bidirectional follower attraction with no statistically detectable change in win probability and only a small, marginally significant decrease in net follower growth. Surface competitive outcomes, absolute financial gains, and follower exchange therefore do not necessarily move in parallel: a lower probability of winning can coexist with gains that the joint performance

generates for the focal streamer. Section 4.2.2 examines which viewers the additional gifting comes from.

4.2.2 Gifting Source Decomposition

We next examine which viewers the additional gifting comes from under each comparison structure. Table 10 shows that, against an opponent with more followers, *Focal Viewer Gifting* rises by 13.31% ($\hat{\beta} = 0.125$, $p < 0.001$) and *Other Viewer Gifting* by 13.88% ($\hat{\beta} = 0.130$, $p < 0.01$), while *Opponent Viewer Gifting* shows no statistically detectable change ($\hat{\beta} = 0.003$). Facing a bigger opponent therefore raises gifting both from the focal streamer’s own pre-PK audience, consistent with the competitive dimension, and from viewers who were in neither streamer’s room before the PK, consistent with the cooperative dimension, in which the jointly produced contest draws in new arrivals. There is no detectable gain from the opponent’s own viewers.

Table 10: Decomposition of the Gifting Response by Viewer Source, Big Follower Opponent (Study 2)

	Focal Viewer Gifting (Log)	Opponent Viewer Gifting (Log)	Other Viewer Gifting (Log)
	(1)	(2)	(3)
vs. Big	0.125*** (0.032)	0.003 (0.005)	0.130** (0.043)
Obs	27,510	27,510	27,510
Controls	Yes	Yes	Yes
Date FE	Yes	Yes	Yes

Notes: Standard errors clustered at the streamer level in parentheses; + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. vs. Big = vs. Big Follower Opponent.

Table 11 presents the analogous decomposition for same-category matchups. *Focal Viewer Gifting* rises by 10.74% ($\hat{\beta} = 0.102$, $p < 0.01$) and *Opponent Viewer Gifting* by 1.26% ($\hat{\beta} = 0.01255$, $p < 0.05$), whereas *Other Viewer Gifting* shows no statistically detectable change ($\hat{\beta} = 0.025$). Same-category pairing therefore raises gifting from the focal streamer’s own audience, again consistent with the competitive dimension, and, to a small extent, from the

opponent’s audience, consistent with the cross-audience exposure through which the cooperative dimension can appear. There is no detectable gain from viewers outside either room.

Table 11: Decomposition of the Gifting Response by Viewer Source, Same Category Opponent (Study 2)

	Focal Viewer Gifting (Log)	Opponent Viewer Gifting (Log)	Other Viewer Gifting (Log)
	(1)	(2)	(3)
vs. Same	0.102** (0.032)	0.01255* (0.00507)	0.025 (0.043)
Obs	27,510	27,510	27,510
Controls	Yes	Yes	Yes
Date FE	Yes	Yes	Yes

Notes: Standard errors clustered at the streamer level in parentheses; + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. vs. *Same* = vs. *Same Category Opponent*.

Taken together, the viewer-source patterns show where the gifting gains under the two comparison structures come from, including the gain that coexists with a lower win probability against a bigger opponent. Under both structures the focal streamer’s own viewers give more, consistent with the competitive dimension, and under both a second audience contributes gifting in a way consistent with the cooperative dimension. Which audience that is differs: against a bigger opponent it is viewers from outside either room, whom the jointly produced contest draws in, whereas against a same-category opponent it is the opponent’s own viewers, whom the split screen exposes to the focal streamer. Comparison structure thus shapes which audiences carry the cooperative gains rather than merely changing the size of the gifting gain.

4.2.3 Heterogeneity by Streamer Characteristics

We next examine whether these effects depend on the focal streamer’s own follower size.

Comparison structure is relative: the same opponent is an upward comparison for a streamer with a small following and a downward one for a streamer with a large following, so the two structures need not mean the same thing to every focal streamer. The audiences differ as well. A small streamer’s viewers are few, so each supporter’s gifts weigh more in the contest score, and the exposure that a joint performance brings is worth more to a streamer whom few viewers

would otherwise encounter; visits from established streamers raise the revenue of emerging streamers (Guo et al., 2025), and audience size itself shapes gifting and engagement in livestreaming (Lu et al., 2021; Zhao et al., 2025). The answer also bears on practice: because only one streamer in a pair faces a bigger opponent, matching is a pair-level allocation; a platform can decide which member of a pair receives the upward comparison, and a streamer can decide which matchups to seek. We therefore categorize focal streamers as ‘Big’ (10,000 followers or more) or ‘Small’ (otherwise) and estimate Equations (5) and (6) (full results in Online Appendix J). Figure 2 plots the predicted average total gifting across focal-opponent configurations, with Panel (a) showing follower-size comparison and Panel (b) showing content-category alignment. Figure 2(a) shows that facing a bigger opponent raises predicted gifting for both small and big focal streamers, but the gain is substantially larger for small streamers (from 2.75 [2.58, 2.92] to 3.25 [3.09, 3.42]) than for big streamers (from 3.29 [3.08, 3.49] to 3.51 [3.29, 3.73]). Figure 2(b) shows that same-category alignment confers a gifting premium on small streamers (from 2.92 [2.76, 3.09] to 3.14 [2.98, 3.31]) but no statistically detectable premium on big ones (from 3.49 [3.29, 3.69] to 3.46 [3.24, 3.67]). Small streamers therefore gain from both a bigger and a same-category opponent, whereas big streamers gain from a bigger opponent and show no statistically detectable gain from category alignment. On the follower side the pattern runs the other way: the new and net follower gains from facing a bigger opponent are larger for big focal streamers (Online Appendix J).

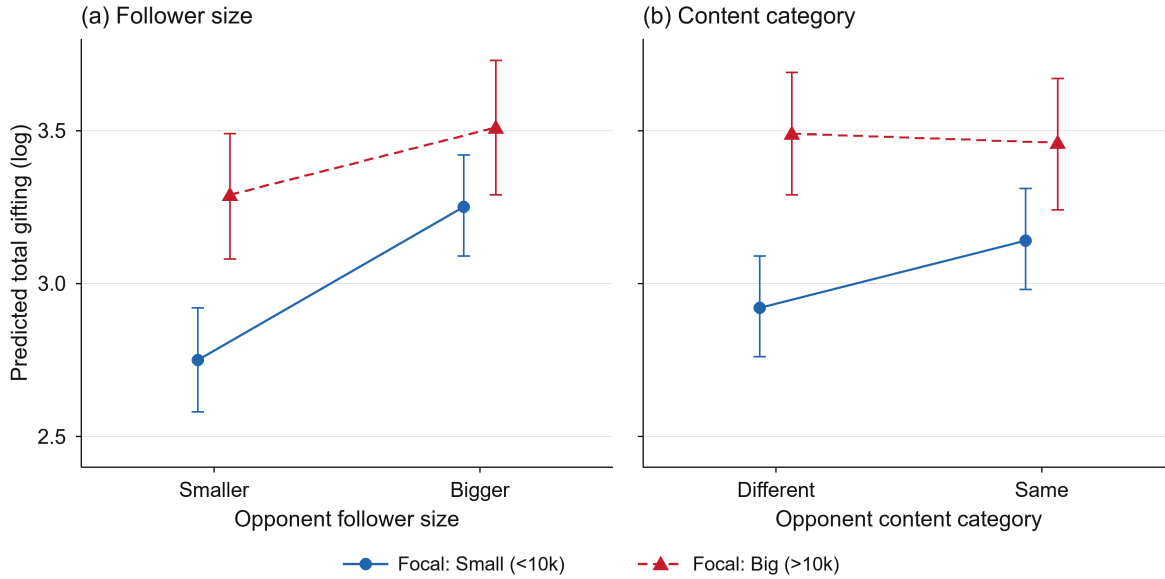


Figure 2: Predicted Average Total Gifting (Log) by Focal Follower Size

Notes: Panel (a): focal $\times$ opponent follower size (coefficient estimates in Online Appendix Table J1, column 1). Panel (b): focal follower size $\times$ opponent content-category alignment (Online Appendix Table J4, column 1). Markers are predicted margins; whiskers are 95% confidence intervals. ‘Big’ focal streamers have 10,000 followers or more. Both panels share a common vertical axis; the horizontal axis is ordered from the lower- to the higher-comparison condition (smaller to bigger opponent; different to same category) to match the baseline-to-treatment reading.

4.2.4 Additional Analysis

We conduct two additional analyses to examine the processes and subsequent outcomes associated with comparison structure. First, we perform a behavioral-engagement mediation analysis using views, likes, comments, and shares during the PK window as parallel mediators. On the mediation subsample, these measures carry 44.2% of the total gifting effect of facing an opponent with more followers and 86.4% of the effect of facing a same-category opponent (Online Appendix K). Second, we examine additional outcomes capturing cross-following, the return of previously inactive viewers, adaptation in the next Random PK session, and the views and loyal-fan gains the focal streamer receives in subsequent Random PK sessions over the following 30 days. Facing an opponent with more followers increases cross-following, returning viewers, and the views and loyal-fan gains in subsequent Random PK sessions, whereas facing a

same-category opponent increases the views in those sessions but reduces returning viewers; the other outcomes show no statistically detectable change (Online Appendix L).

In summary, Study 2 shows that PK outcomes can diverge across dimensions and that comparison structure shapes the audience responses behind them. Facing an opponent with more followers lowers the focal streamer's probability of winning while increasing total gifting and net follower growth, so a lower probability of winning coexists with financial and audience gains. Facing a same-category opponent produces no statistically detectable change in win probability but increases total gifting and bidirectional follower attraction. The viewer-source analysis shows a pattern consistent with these gains arising from the joint audience engagement and discovery of PK: under either structure the focal streamer's own viewers give more, and a second audience contributes as well, viewers from outside either room against a bigger opponent and the opponent's own viewers against a same-category opponent. Heterogeneity analyses further show that the gain from a bigger opponent is larger for small streamers, and that the gain from a same-category opponent appears for small streamers, with no statistically detectable gain for big ones.

5. DISCUSSION

5.1 Key Findings

This study investigates how PK, as a form of coopetitive social comparison, affects streamer performance. Study 1 shows that PK participation substantially increases gifting, viewer engagement, and follower acquisition. A decomposition of total gifting shows that PK increases viewer count and average gifting per gifter with no statistically detectable change in the gifting conversion rate; these two margins are consistent with the cooperative and competitive dimensions of PK. A mediation analysis of viewer comments shows a significant indirect association through competition awareness, reflected in comments referring to the contest, consistent with the competitive aspect of coopetitive social comparison. Study 2 shows that PK

outcomes can diverge across dimensions: facing a bigger opponent lowers the focal streamer's likelihood of winning but increases gifting and net follower growth, while same-category matching increases gifting and bidirectional follower attraction with no statistically detectable change in win probability. These patterns show how competitive losses can coexist with gains generated by the joint audience engagement and discovery of PK: under both structures the focal streamer's own viewers give more, and a second audience contributes as well, namely viewers outside either pre-PK audience against a bigger opponent and the opponent's own viewers against a same-category opponent. Comparison structure thus shapes which audiences the gains come from. Together, the findings show that competition and mutual gains coexist in PK, and that comparison structure shapes the audience responses through which these cooperative outcomes unfold.

5.2 Implications for Research

Our work offers several implications for research. For social comparison theory, the findings locate the behavioral effects of comparison in the audience rather than only in the comparer, and they extend the comparison structures the theory considers. Prior research has predominantly examined social comparison as a comparer-centered process in which individuals evaluate themselves and adjust their own emotions or behaviors (Krasnova et al., 2015; Li et al., 2021; Liu et al., 2017; Shi et al., 2025). When a comparison is externalized onto a public stage, as in PK, it triggers a compare-choose-support chain among third-party viewers, whose collective behavior determines the comparison outcome; the mediation analysis of viewer comments is consistent with this reading, since the indirect association runs through comments about the contest rather than through expressions of excitement or affiliation.

Comparison structure then determines the composition of the audience's gifting response. Follower-size comparison and content-category alignment, the latter a structure beyond the comparison direction that organizes existing work, shape which audience segments respond: the focal streamer's own supporters give more under both structures, whereas the second responding audience is viewers outside either pre-PK audience when the opponent has more followers and the opponent's own viewers when the opponent streams in the same category. The heterogeneity results add that the same comparison structure produces a larger gifting response when the focal streamer is small, so the comparer's own standing moderates the audience-activated effect. Comparer-centered studies of a single comparer's reaction leave both the composition of the response and this moderation open.

For coopetition theory, PK shows that how competition and cooperation are expressed within a single interaction depends on its comparison structure. Prior work has examined how competition and cooperation coexist in coopetitive relationships (Hoffmann et al., 2018), but less is known about how features of the interaction structure shape the way each is expressed. Coopetitive social comparison yields concurrent outcomes: a win-lose result can coexist with absolute financial gains for both participants (Study 1) and an exchange of followers between them (Study 2), so evaluating coopetition requires attention to multiple performance dimensions rather than a single relative outcome. Which audience responses carry the coopetitive gains also depends on comparison structure: the supporter response consistent with the competitive dimension appears under both structures, whereas the response consistent with the cooperative dimension comes from new arrivals under an upward comparison and from the opponent's own audience under category alignment. The expression of competition and cooperation in a coopetitive interaction is therefore not fixed but depends on the comparison structure the

platform sets, which makes comparison structure a design choice that can promote engagement and monetization.

For gamification research, our results provide evidence on an integrated gamified design in a synchronous, collective setting. Prior research has predominantly examined individual gamification elements separately and in asynchronous, individual-focused environments (Ho et al., 2023; Liu et al., 2017; Wang et al., 2026). PK combines real-time competition, dynamic scoring, time pressure, audience participation, and reward-punishment into one experience, and the design works not only by motivating an isolated individual but by mobilizing an audience whose collective engagement determines the competitive outcome. The results show how such a design can operate in a live, collective environment to shape both immediate engagement and subsequent streamer outcomes.

5.3 Implications for Practice

Our results provide actionable implications for digital platform operators and content creators. Platforms without real-time competitive features can consider implementing PK-style cooperative social comparison as an effective way to drive user engagement and monetization. For platforms already offering PK functionality, matching is a pair-level allocation: in any pair only one streamer faces a bigger opponent, so a platform chooses which side receives that condition rather than granting it to everyone. The heterogeneity analyses indicate where the gain is largest: the gifting gain from facing a bigger opponent is larger for small streamers, which is an argument for allocating the upward comparison to them, and same-category pairing raises gifting for small streamers, with no statistically detectable effect for big ones. Although such matchups may appear unbalanced, our results show that smaller focal streamers receive more gifts and gain more followers when facing larger opponents than when facing smaller ones, despite being less

likely to win the contest. In addition, platforms can implement incentive mechanisms, such as social nudges or targeted reputational rewards, to encourage big streamers to participate in PK events with smaller peers, thereby promoting a more inclusive and balanced creator ecosystem. For streamers with small follower bases, both a bigger opponent and a same-category opponent are associated with higher gifting, so either matchup may be beneficial; for big streamers, only a bigger opponent is associated with higher gifting.

5.4 Limitations and Future Research

This work has several potential limitations, which also point to fruitful directions for future research. First, streamers choose whether to participate in PK events, so the Study 1 estimates rest on conditional independence and common support given the observables; the matching-based estimates and balance diagnostics in Online Appendix C address the comparability of sessions with and without PK on those observables, while time-varying factors we do not observe, such as a streamer's mood or expectations for a given session, remain a possible source of selection. Because a PK requires two streamers to agree to participate in the first place, an experiment can only randomize eligibility for the feature at the platform level rather than assign participation directly; such a design, if ethically feasible, is a next step. Second, our two studies investigate the immediate effects of PK and those over the following 30 days. Future research can explore the longer-term effects of repeated participation. It is possible that the benefits of competition diminish over time, or that repeated losses could have a negative impact on a streamer's motivation and community. Third, though we focus on the streamers and streamer-side outcomes, it would be useful to also understand viewer-side perspectives in future research. Examining streamer-viewer interactions during PK events would extend this line of research. Fourth, our analysis centers on a single livestreaming platform in Asia. The framework is most

likely to apply in settings that share PK's four structural features: a paired and publicly displayed comparison, an audience whose contributions decide the outcome, real-time interaction, and a bounded contest window, as in TikTok Live; it may not extend to asynchronous settings such as YouTube. Further studies across platforms and cultural contexts would strengthen external validity.

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